

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (CSE/ISE/AIIML/CCE) (Credit System) Degree Examinations**

April 2023

21CS302 / 20CS305 / 21IS302 / 21AM302 / 21CC302 – DATA STRUCTURES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain memory management functions used in C for dynamic memory management.	6	L*2	1	1
	b) With a suitable example explain Array of structure and Array of pointers.	4	L2	1	1
	c) Write C functions to read, store and display Student name, reg_no, marks scored in 5 subjects by n students as structure members and find the total marks and display the same.	10	L3	1	1,3
2.	a) What is a stack? Explain working of a stack giving it's primitive operations with stack overflow and stack underflow conditions.	6	L2	2	3
	b) Convert the following infix expressions to prefix and postfix expressions: i) $((A+B)*C-(D-E))(F+G)$ ii) $A\$B*C-D+E/F/G+H$	4	L3	2	1
	c) Give an algorithm and corresponding C function to evaluate a given valid postfix expression and show the traces of evaluation of $6\ 2/\ 2-42*+$ with the contents of stack.	10	L3	2	3
3.	a) Give an algorithm to convert a valid arithmetic infix expression to its equivalent postfix expression.	6	L2	2	2,3
	b) Convert the following infix expression to postfix using conversion algorithm $A*(B+C)/D$, showing contents of stack in each stage.	4	L3	2	3
	c) What is recursion? Write recursive function to compute factorial of a positive number and trace it for computing factorial of 5.	10	L3	2	2,3
Unit – II					
4.	a) What is a Queue? Explain in brief the same. Design C functions for inserting and deleting an element to and from a circular queue.	6	L2	3	2
	b) Discuss the drawback of ordinary queue. Give two methods used to overcome it?	4	L2	3	1
	c) What is a Linked list? Write C functions to: i) Insert after a node whose data value is given ii) Delete a node from the end of list iii) Count the number of nodes in a linked list	10	L3	4	3
5.	a) Write a C functions - insert and delete to represent a linked list as a queue.	6	L3	4	3
	b) Write a C function to search an element in a linked list.	4	L3	4	3
	c) Write C Functions to i) insafter(p,x) and delafter(p,x) to insert and delete a node after a node with node pointer p respectively in a circular singly linked list. ii) Concatenate two circular singly linked list.	10	L3	4	3

P.T.O.

6. a) Give a suitable data structure to create a doubly linked list. Write C function to insert a new node from the front of the Doubly linked list. 6 L3 4 2
- b) Develop a C function to delete all the occurrences of a given value from the singly linked list. 4 L3 4 3
- c) Develop C routines:
- i) Insert a given value into an ordered doubly linked list at its proper positions. 10 L3 4 1,3
- ii) Given a singly linked list of integers, create two separate lists for odd and even integers respectively.

Unit – III

7. a) Define the following with supporting diagram.
(i) Binary Tree (ii) Skewed binary tree (iii) Complete binary tree
(iv) Fully binary tree. 5 L2 5 1
- b) Construct the binary tree from the following array representation of binary tree:
- | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|
| A | B | C | D | - | E | - | G | - | - | - | F | -- | - | - |
|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|
- Give the formula to determine the left and right child of a node in the array representation of a tree. 5 L3 5 2
- c) Define a binary Search tree (BST)
- (i) Create BST for the following data {35,20,30,45,55,40,38,18,22}
- (ii) Write C routines for inorder, postorder and preorder traversals.
- (iii) Write the inorder, preorder, postorder traversals for the tree constructed in (i). 10 L3 5 3
8. a) Write an expression tree to represent an expression $A+B^*C-D/E$ and give the result of preorder and post order traversals on that tree. 6 L3 5 3
- b) What do you mean by collision in hashing? Show how the collision resolution is made using linear probing technique in hashing for Record: A B C D E X Y Z
H(K): 4 8 2 3 4 2 7 9 of size 10. 4 L3 5 2,3
- c) Explain two ways of representing the given undirected graph G(5,6) giving the significance of the two methods.
Graph G: Nodes: ABCDE
Edges: AB, BC, CD, AD, BE, EC 10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
